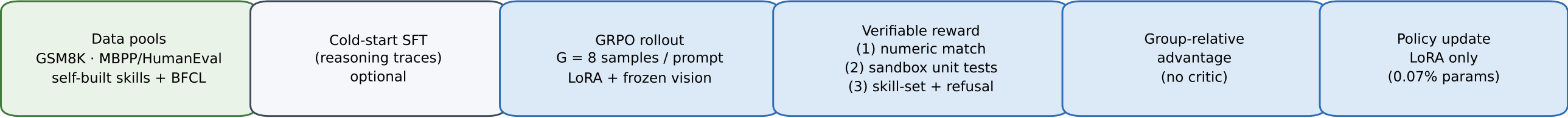


(a) One shared pipeline over three verifiable tasks



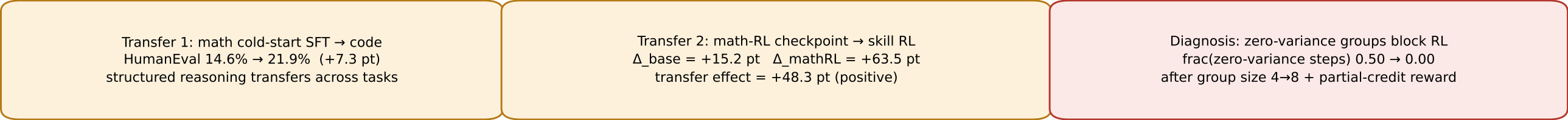
on-policy loop (resumable, checkpointed)

Single GPU: 12 GB consumer (4-bit/LoRA) or 32 GB V100 · SFT and RL question pools are disjoint (no leakage) · training pools self-built, evaluation on public benchmarks

(b) Three capability layers (same recipe, different verifier)

Reasoning	GSM8K (answer matching)	34.0% → 50.0% (base 30.0%)	surpasses the 9B baseline (40.0%)
Execution	HumanEval / MBPP (sandboxed tests)	21.9% → 25.0% (base 14.6%)	custom sandbox: timeout · no-net · rlimits
Selection	BFCL v3 (skill set + refusal)	0.795 exact (retrieval 0.485 / rule 0.590)	false-load rate 100% → 22.5%

(c) Cross-layer transfer and the key diagnosis



Metrics: GSM8K = exact accuracy on 50 test items (full-suite run pending) · HumanEval = pass@1 over 164 tasks · BFCL v3 = stratified 400 items (80 per category); false-load = loading a skill when none is required.

All results are reproducible from this repository (see FINAL_REPORT.md §6); curated raw evidence (result JSONs, training logs) is in evidence/.